

思考 8: b

(1) $\lim_{z \rightarrow \infty} \psi(z) \psi(z) = \infty$

是 $(m+n)$ 级极点

(2) $\lim_{z \rightarrow \infty} \psi(z) - \psi(z)$ 无穷

是 $\max\{m, n\}$ 级极点

(3) ① 若 $m > n$ 无穷

$(m-n)$ 级

② 若 $m \leq n$ $\lim_{z \rightarrow \infty} \frac{\psi(z)}{\psi(z)} = 0$ 可去

1(1) $z=0$ 单极点

$z=\pm i$ 二阶极点

1(2) $z^4 = -1, z = e^{i\theta}$

$e^{i4\theta} = e^{i\pi}$

$4\theta = \pi + 2k\pi$

$\theta = \frac{\pi}{4} + \frac{1}{2}k\pi, z_1 = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}, z_2 = -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}, z_3 = -\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}, z_4 = \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}$

均单极点

1(7) $z=1$ 二阶极点

1(b) $e^{\frac{1}{z-1}}, z=1$

$e^{\frac{1}{z-1}} = \sum_{n=0}^{\infty} \frac{(\frac{1}{z-1})^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} (z-1)^{-n}$ 本性奇点

2(z) $z = \frac{\pi}{2} + k\pi$ 可去奇点

$\text{Res}[\tan z, \frac{\pi}{2} + k\pi] = 0$

(4) $z=0$ 二阶极点

$\text{Res}[\frac{1+e^z}{z^2}; 0] = \left(\frac{1+e^z}{z^2} \right)' \Big|_{z=0} = \frac{e^z z - z^2(1+e^z)}{z^4} \Big|_{z=0} = \frac{z^2 - 2z(1+e^z)}{z^4} \Big|_{z=0} = \frac{(z - 2 - 2e^z)}{z^3} \Big|_{z=0} = -2$

$\text{Res}[\frac{1}{z}; 0] = 1$

$\therefore \text{Res}[\frac{1+e^z}{z^2}; 0] = 2$

2(5) $z=2$ 本性奇点

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\sin \frac{1}{z-2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(z-2)^{2n+1}} \quad \text{当 } n=0 \text{ 时}$$

$$\text{Res}[\sin \frac{1}{z-2}; 2] = \text{coefficient of } \frac{1}{z-2} = 1, \quad \text{LHS} = 2\pi i$$

3(1) $z=0$, 可去奇点

$$\text{Res}[\frac{\sin z}{z}, 0] = 0, \quad \therefore \oint_{|z|=\frac{3}{2}} \frac{\sin z}{z} dz = 0$$

$$(3) \quad z^2 + 2z + 5 = 0$$

$$z = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i \quad \text{不在区域}$$

$$\text{Res}[\frac{z}{z^2 + 2z + 5}; -1 \pm 2i] = \frac{z}{2z+2} \Big|_{-1 \pm 2i} = \frac{2+i}{4} \text{ 及 } \frac{2-i}{4}$$

$$\therefore \text{LHS} = 2\pi i \left[\frac{2+i}{4} + \frac{2-i}{4} \right] = 2\pi i \quad \text{D}$$

(6) $z=0, \pm 2\pi i$ 单极点

$$\text{Res}[\frac{1}{e^z - 1}; 0] = \frac{1}{e^z} \Big|_{z=0} = 1$$

$$\text{Res}[\frac{1}{e^z - 1}; 2\pi i] = \frac{1}{e^z} \Big|_{z=2\pi i} = 1$$

$$\text{Res}[\frac{1}{e^z - 1}; -2\pi i] = \frac{1}{e^z} \Big|_{z=-2\pi i} = 1$$

$$\therefore \text{LHS} = 2\pi i \cdot 3 = 6\pi i$$

法2: $f_1(z) = \frac{p_1(z)}{z-z_0}$

$f_1(z)f_2(z) = \frac{p_1(z)p_2(z)}{(z-z_0)^2} = \frac{p_1(z)p_2(z)}{(z-z_0)^2}$

$f_2(z) = \frac{p_2(z)}{z-z_0}$

$= p_1(z)p_2(z) + p_2'(z)p_1(z)$

$\text{Res}[f_2(z), z_0]$

法1: $f_1(z) = a_{-1}(z-z_0)^{-1} + \sum_{n=0}^{\infty} a_n(z-z_0)^n$

$f_2(z) = b_{-1}(z-z_0)^{-1} + \sum_{n=0}^{\infty} b_n(z-z_0)^n$

$\therefore f_1(z)f_2(z) = a_{-1}b_{-1}(z-z_0)^{-2} + \dots$ 为二级极点

$\therefore \text{Res}[f_1(z)f_2(z); z_0] = a_{-1}b_{-1} = \text{Res}[f_1(z); z_0] \text{Res}[f_2(z); z_0]$

不对的, 这是(-2)项, 要求(-1)项

7 (1) $z=0$, 单极点; $z=-2$, 三级极点

$\text{Res}\left[\frac{(z-1)^3}{z(z+2)^3}; 0\right] = \left.\frac{(z-1)^3}{(z+2)^3}\right|_{z=0} = -\frac{1}{8}$

$\text{Res}\left[\frac{(z-1)^3}{z(z+2)^3}; -2\right] = \frac{1}{2!} \left.\left(\frac{(z-1)^3}{z}\right)''\right|_{z=-2} = \frac{1}{2} \cdot \frac{4 \times 36 + 2 \times 9(-3)}{16} = \frac{9}{8}$

$\therefore \text{LHS} = 2\pi i \left[-\frac{1}{8} + \frac{9}{8}\right] = 2\pi i$

8 (2) ~~$\cos \theta$~~ $z = e^{i\theta}$ $dz = ie^{i\theta} d\theta$ $d\theta = \frac{1}{iz} dz$

$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + \frac{1}{z}}{2} = \frac{z^2 + 1}{2z}$

$\text{LHS} = \frac{1}{2} \int_{-1}^1 \frac{1}{2 \cos \theta + 3} d\theta = \frac{1}{2} \oint_{|z|=1} \frac{1}{\frac{z^2+1}{z} + 3} \frac{1}{iz} dz = \frac{1}{2i} \oint_{|z|=1} \frac{dz}{z^2 + 3z + 1}$

$z_0 = \frac{-3 \pm \sqrt{9-4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$ 两个极点 $\frac{3-\sqrt{5}}{2}$ 不在 $|z|=1$ 内

$\therefore \text{Res}\left[\frac{1}{z^2 + 3z + 1}, \frac{-3 + \sqrt{5}}{2}\right] = \left.\frac{1}{2z+3}\right|_{\frac{-3+\sqrt{5}}{2}} = \frac{1}{\sqrt{5}}$

同理 $\text{Res}\left[\frac{1}{z^2 + 3z + 1}, \frac{-3 - \sqrt{5}}{2}\right] = -\frac{1}{\sqrt{5}}$

$\therefore \text{LHS} = \frac{1}{2i} \cdot 2\pi i \left[\frac{1}{\sqrt{5}} - \frac{1}{\sqrt{5}}\right] = 0 = \frac{\pi}{\sqrt{5}}$